CollegeCopilot - Complete Project Summary

For context handoff | GitHub: Gunar-Oguz/College-Copilot

WHAT THE PROJECT IS

CollegeCopilot is an AI-powered college search tool for admissions counselors.

* Backend: FastAPI (backend/main.py) - /ask endpoint turns plain English into SQL
* Database: PostgreSQL with College Scorecard schools + programs data
* Frontend: Chat UI at frontend/index.html
* Repo: https://github.com/Gunar-Oguz/College-Copilot
* Local URL: http://localhost:8000 (after starting uvicorn)
* Deploy: render.yaml for Render.com (no Dockerfile)

PROJECT FILE TREE

```
college_cop/
  .env                          API keys (local only, gitignored)
  backend/
    main.py                       FastAPI + /ask endpoint
    database.py                   PostgreSQL connection
    query_parser.py               Natural language -> filters
    sql_builder.py                Filters -> SQL
    query_engine.py               Full query pipeline
    response_formatter.py         Format chat responses
    data_coverage.py              Data coverage stats
    load_all_schools.py           Load all schools (single year)
    load_all_years.py             Load schools 1996-2024
    load_recent_years.py          Load schools 2015-2024
    load_programs.py              Load programs/majors data
    requirements.txt
  data/
    schema.sql                    Schools table (37 columns)
    programs_schema.sql           Programs table schema
  frontend/
    index.html                    Chat UI
  tests/
    test_queries.py               44 query engine tests
    test_load_programs.py         Programs loader tests
    test_load_all_years.py        Schools earnings loader tests
  render.yaml                     Deploy config
  README.md
```

PHASE 1: CORE APP (DONE)

* Built FastAPI backend with /ask endpoint
* Query engine: natural language -> SQL (query_parser.py, sql_builder.py)
* Built chat frontend (HTML)
* 44 tests in tests/test_queries.py

PHASE 2: SCHOOLS DATA (DONE)

* Created data/schema.sql - schools table with 37 columns
* load_all_years.py - load 1996-2024 (hit rate limits on early years)
* load_recent_years.py - load 2015-2024 only (~39 min, more reliable)

* Multiple years of school data loaded into PostgreSQL

PHASE 3: PROGRAMS DATA (DONE + FIXED)

* Created data/programs_schema.sql - programs table with FK to schools
* Created load_programs.py - Field of Study data 2015-2024
* BUG FOUND: All earnings columns were NULL
* CAUSE: Loader used earnings.2_yr but API has 1_yr, 4_yr, 5_yr (not 2_yr)
* FIX: Changed to earnings_median_1yr and earnings_median_4yr columns
* FIX: Extract earnings.1_yr.overall_median_earnings and earnings.4_yr.overall_median_earnings
* RESULT: ~1,840,795 programs loaded across 9 years
* Years: 2015-16 through 2024-25 (2023-24 skipped - no API data)
* Added retry logic for ChunkedEncodingError and ReadTimeout
* Added --schema-only flag to recreate table without API key

PHASE 4: SCHOOLS EARNINGS FIX (DONE)

* BUG: earnings_median_10yr was NULL for all years except 2020-21
* CAUSE: For 2024-25 API uses latest.earnings.10_yrs_after_entry.median
* Historical years use {year}.earnings.10_yrs_after_entry.median
* FIX: Added api_prefix() - returns 'latest' for 2024, year string for historical
* FIX: Updated fields_for_api_year() and _get() to use correct prefix
* BUG: online_only column is boolean but API returns 0 or 1
* FIX: Added _as_bool() to convert 0/1 to True/False before insert
* Added --recreate flag to load_all_years.py
* Test plan: reload 2024-25 only first, verify COUNT, then reload all years

GIT / GITHUB

* All fixes committed and pushed to main branch
* Repo: https://github.com/Gunar-Oguz/College-Copilot
* Local Mac synced with origin/main

ENVIRONMENT VARIABLES (.env)

```
ANTHROPIC_API_KEY=your_key
COLLEGE_SCORECARD_API_KEY=your_key
DATABASE_URL=your_postgres_url  (optional, defaults to local postgres)
```

KEY COMMANDS

```
# Run the app
cd backend
uvicorn main:app --reload --host 127.0.0.1 --port 8000
# Open http://localhost:8000

# Load schools - test one year
python load_all_years.py --start-year 2024 --end-year 2024 --recreate

# Load schools - recent years with full schema
python load_recent_years.py --recreate

# Load programs
python load_programs.py --recreate

# Run tests
python -m pytest ../tests/test_queries.py -v
```

```
# If port 8000 busy:
lsof -ti :8000 | xargs kill -9
```

VERIFY QUERIES (SQL)

```
-- Schools earnings test (2024-25)
SELECT COUNT(earnings_median_10yr)
FROM schools WHERE year = '2024-25';
-- Expected: ~5000+ (not 0)

-- Programs earnings by year
SELECT year,
       COUNT(earnings_median_1yr) AS has_1yr,
       COUNT(earnings_median_4yr) AS has_4yr
FROM programs
GROUP BY year ORDER BY year;
-- Expected: non-zero counts per year

-- Example program query
SELECT s.name, p.program_name,
       p.earnings_median_1yr, p.earnings_median_4yr
FROM schools s
JOIN programs p ON s.id = p.school_id AND s.year = p.year
WHERE p.program_name LIKE '%Computer Science%'
  AND s.year = '2024-25'
ORDER BY p.earnings_median_4yr DESC
LIMIT 10;
```

PROGRAMS DATA BY YEAR

```
Year       Programs
2024-25    216,684
2022-23    216,551
2021-22    197,613
2020-21    211,491
2019-20    212,119
2018-19    205,163
2017-18    198,844
2016-17    194,012
2015-16    188,318
Total:     1,840,795 programs
```

NOT DONE YET

* Full schools reload with all 37 columns for all years
* Year selection in query parser (currently hardcoded to 2024-25)
* Wire programs into /ask for program-level queries (e.g. CS earnings)
* Dockerfile not created

CURRENT STATUS

```
Item                          Status
----------------------------------------
App + query engine              DONE
Programs data (~1.84M rows)     DONE (earnings fixed)
Schools earnings fix (code)     DONE, on GitHub
Schools 2024-25 test reload     User ran - verify COUNT
Full multi-year schools reload  PENDING
Program queries in chat         NOT WIRED
Dockerfile                      NO
```